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Hybrid vertical takeoff and landing unmanned aerial vehicle



Team: ASIANS Topic: 2 Date: 06.12.2025

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Concept selection & Advantage Justification

UAV Key Design Details:

Design 1

Drone Layout: Hybrid configuration (rotor + fixed-wing)
Combines vertical takeoff advantage of rotors & long-endurance/long-distance merits of fixed wings, meeting 10kg cargo transport over 5km in complex urban environments

Design 2

Structural Materials: Mainly carbon fiber composites, minor metal structures
Carbon fiber offers far higher specific strength/stiffness than aluminum, also fatigue-resistant and corrosion-resistant for complex flight environments



Design 3

Control Scheme: 3 motors + 2 ailerons (no rudder/elevator to reduce weight & boost effective load ratio)

- Pitch: Controlled by tail motor
- Roll: Controlled by ailerons
- Yaw: Controlled by motor differential & roll

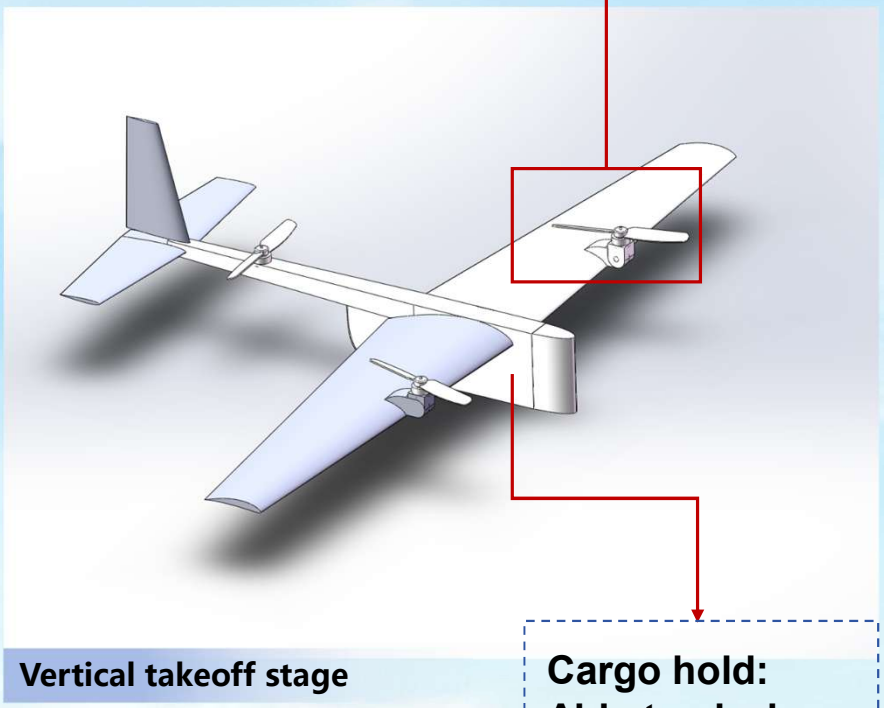
Design 4

Transport Scheme: Vertical takeoff with load → Airdrop load at designated area → Return to base

Conceptual Appearance

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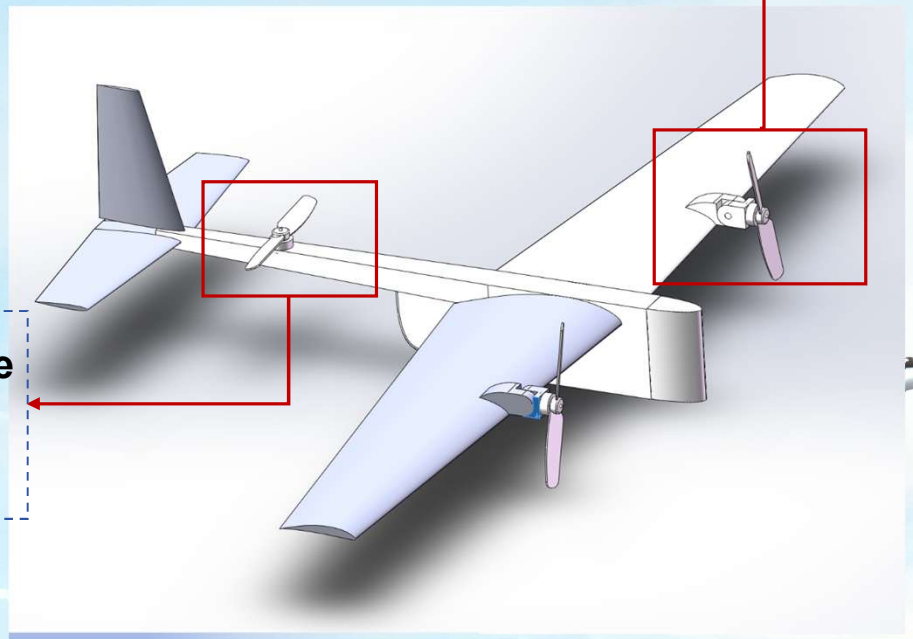
The two motors on the wing can change the direction of the thrust



Vertical takeoff stage

Cargo hold:
Able to air-drop

Control the pitch attidude



Long-distance flight stage

Conceptual Appearance

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Takeoff video display



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Company Workflow

Establish a three-level security management framework:

3 Principle

- Decision making level (safety committee): Develop safety standards and approve flight plans;
- Execution layer (flight control/operation and maintenance team): Implement operational standards and perform equipment inspections;
- Supervisory layer (safety inspector): Real time monitoring of flight and post audit of hidden dangers.



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Company Workflow

emergency priority principle

equipment level： each drone is equipped with single battery hot standby and flight control system backup

Operational level: pre-set "emergency landing point+emergency contact person"

Training level: Supervisors need to pass a "fault simulation assessment" (such as motor failure and signal interruption).



Company Workflow

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Charging safety regulations:



- Fire extinguishing equipment should be equipped in the charging area and kept away from heat/fire sources
- Battery life tracking: Mandatory retirement after 200 cumulative charge and discharge cycles.



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Cost

Type	Name	Quantity	Price (USD)
Structural Material Consumables	Carbon Fiber Composite Material	Large Quantity	850
	Metal	Small Quantity	140
	General Consumables	Several	110
Power & Energy System	Motor + Propeller	3	360
	Battery	1	600
Electronic Control System	Flight Control Board	1	425
	GPS, Airspeed Indicator, etc.	Several	210
Actuator	Servo Motor	3	170
	Rotating Device	2	140
	Total Price	-	3005



Calculations

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Drone Design Parameters
Gross Weight: 20 Kg
Airfoil: MH115
Cl / Cd = 90
V = 30 m/s
E_required for 1 round trip = 20 KJ = 5.55 Wh
Flights per change = 10 Round Trips
Total E_required = 55.55 Wh

Weight per parameter

1 Battery	3 kg
3 Motors	3.6 kg
4 Servos	2 kg
1 Flight Controller	
Sensors	

Design Goals
1. 5 km radius Range
2. 10 kg Payload

Flight Controller

Axisflying Argus Pro

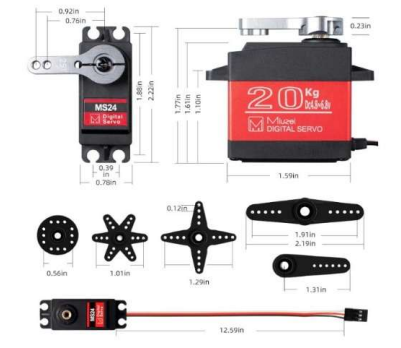




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Calculations





Battery Specifications

Weight:	3 Kg
Type:	Li-ion Battery
Rated:	22.2 V – 36500 mAh
Energy:	600 Wh
Cost:	USD 600



Motor Specifications

Number of Motors:	3
Power Specific Thrust:	4 g/W
Weight:	1.2 Kg

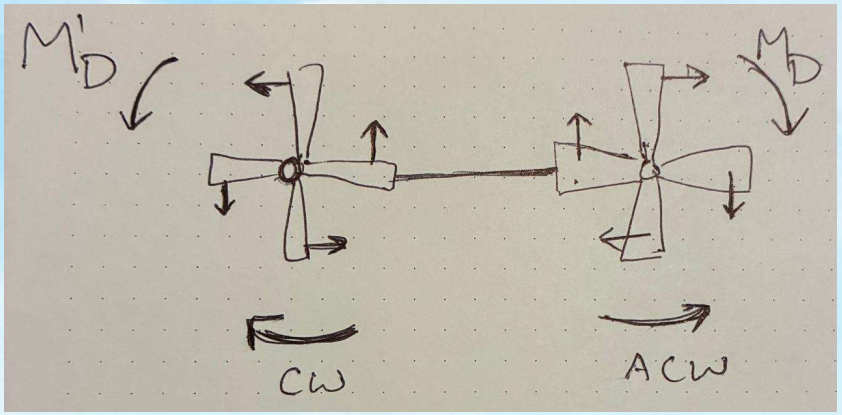
Servo Motor Specifications

Number:	4
Torque:	0.2 Nm
Weight:	0.5 Kg

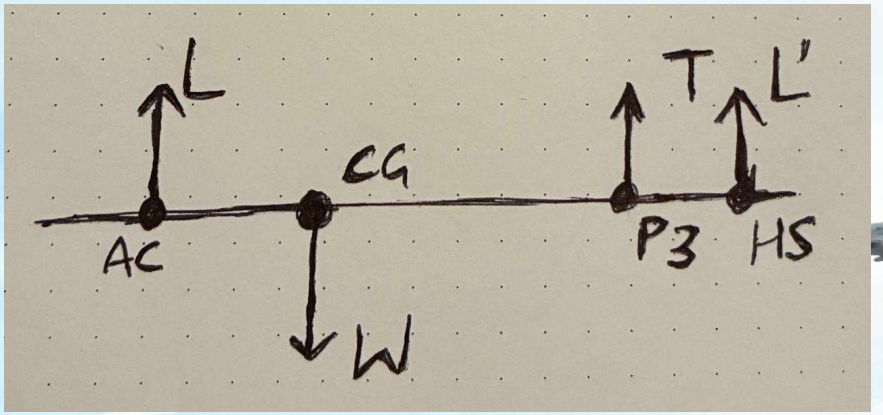
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Force Diagrams

Counter Rotating Propellers

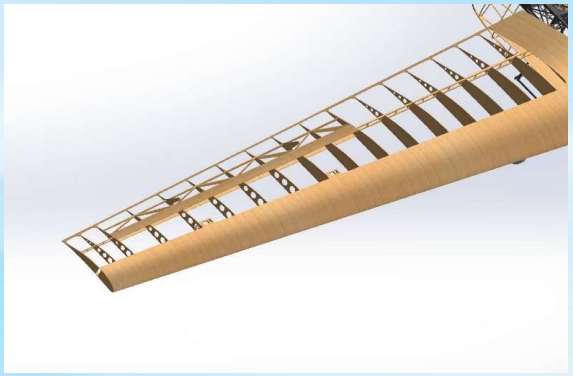


Vertical Force Diagram



Future Outlook

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1. **Detailed structural design** (e.g., load-bearing optimization, aerodynamic refinement)

2. Comprehensive **autonomous flight control system design**

3. Rotating mechanism **selection & design** for the power system

4. Build a physical prototype + conduct flight verification tests

5. **Iterative optimization** based on flight test results





СПАСИБО!

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